



An integrated framework for the assessment of environmental sustainability in wood supply chains

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ABSTRACT

Nowadays, sustainability is one of the critical factors for the success of supply chains in organizations and firms that strive to maintain a competitive edge in the market. In wood industry, due to the need for integrating several business units such as forest entrepreneurs, carriers, pulp and paper mills and sawmills, such industries encounter various difficulties in maintaining effective supply chain collaborations. Literature on wood furniture industry seems to be lagging in terms of research on sustainable supply chain operations. To fill this gap, this study aims at identifying the critical factors that stand as a barrier between manufacturing and environmental sustainability in wood furniture industries. To achieve this aim, an integrated framework based on the triplet of Hierarchical Clustering, Analytical Hierarchy Process and Best-Worst Method has been proposed and implemented in a leading furniture manufacturer in UAE. The results show that waste management is the primary concern when an organization wants to pursue manufacturing environmental sustainability. Resources come in the second place where non-renewable resources should be substituted by renewable ones. The results were supported by a sensitivity analysis which confirms that higher attention should be directed to recycling of wood waste. Findings of this study provide recommendations to managers and decision makers on how to improve the manufacturing environmental sustainability in wood furniture industries to achieve the Triple Base Line (TBL) concept of sustainability and integrate sustainably in industry 4.0 context.

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1. Introduction

The world's growing population and extensive use of fossil-based goods have caused the globe to overstep its planetary bounds, posing a number of social, economic, and environmental issues. For this reason, governments have been promoting green energy, organic and wood-based products over the past few decades as a result of social and political

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pressure to meet climate mitigation targets (e.g. Kyoto Agreement, Paris Agreement). This resulted in fast expanding demand for wood and increased competition for logs and sawmill wood by-products worldwide (Antonio et al., 2021).

Governments are driving a shift towards a low-carbon future via imposing effective strategies and policies to manage emissions and plastic pollution. Likewise, businesses are responding by developing methods to improve their sustainability. With such policies, producers must optimize their systems and focus the efforts on fostering bio-economy, as well as ecofriendly goods and markets to increase the environmental performance as demand for environment low-impact products grows (Alanya-Rosenbaum et al., 2021; Maxim et al., 2021).

One of the most commonly exploited natural resources for home products is wood. Owing to its importance and use in many applications, wood industry is considered vital in modern societies. According to literature studies in 2018, China is ranked first in the list of wood exporters across the world recording a value of USD 13,544,087 thousand US dollars from wood exports. After China, Canada and the United States of America come next followed by Germany and Russian Federation (<https://www.exportgenius.in>, 2018).

The fact of the matter is, the world's forest acreage is quickly shrinking, not just because of the usage of wood for energy, but also because of rising urbanization. Forest area has declined from 31.8% to 30.8% since the beginning of the past century (Peric et al., 2020). In the US for instance, the forest industry accounts for over 4% of total manufacturing GDP in the United States, with yearly forest product output totaling \$210 billions.

In wood industry in particular, design and drafting play an important role in maximizing wood utilization, waste reduction and recycling. Now days, IKEA alone consumes one percent of the world's entire commercial wood supply. As for the Middle East, the imports exhibit some variations from country to country. Although the Middle East is advancing gradually in its efforts towards recycling, there is no significant evidences of efforts undertaken in wood waste and wood by-products recycling (Zafar, 2018). A key concern for the forest industry in these times is how to produce and recycle wood in an efficient and cost-effective manner while also meeting consumer demands (Jigeng et al., 2020). One idea is to support wood supply chains to facilitate the recycling and to better manage the waste. Supply chain management (SCM) of wood has gotten a lot of attention from academics and practitioners in recent years (Sunil and Subrahmanya Bhat, 2021).

What we realized is that research on sustainable supply chain operations, particularly for specialized industries, is underperforming in the UAE and there is no framework that has been suggested to describe and analyze wood supply chain characteristics for furniture industry in UAE. This also highlights the research questions that we aim to answer in this study, *first*, how to identify the factors that are essential for environmental sustainability in wood supply chain, particularly in UAE, *second*, how to construct a new framework based on the integration of Hierarchical Clustering, Analytical Hierarchy Process, and Best Worst Method (HC-AHP-BWM) to assess the factors of environmental sustainability in wood industry, *third*, what strategic recommendation can be incorporated to improve environmental sustainability in wood supply chain.

To close this gap, we aim at the sustainability in the setting of the emerging factors, namely, environmental, economical and social. In particular, given the contexts of the above discussion, a gap in the existing literature could be highlighted, to fill in the gap, the following objectives were considered in this study:

- Identify the factors that are essential for environmental sustainability in wood supply chain.
- Develop an integrated framework based on the triplet of Hierarchical Clustering, Analytical Hierarchy Process, and Best Worst Method (HC-AHP-BWM) to assess the factors of environmental sustainability and to keep them in line with their weights using the developed framework.
- Proposed strategies to improve environmental sustainability in wood supply chains.

In order to achieve the above-mentioned objectives, the study was performed and structured as follows. In Section 2, a literature review of sustainable wood industry and the metrics of sustainability assessment are presented. In the same section, wood supply chain sustainability along with the different multi-criteria decision-making methods are identified. The proposed methodology is discussed in Section 3 followed by a real world application of the proposed framework in Section 4. The results are presented and discussed in Section 5. Finally, the conclusion and suggestions for future research are demonstrated in Section 6.

2. Literature review

In this section, we first start by reviewing sustainability in general, then the metrics and criteria used for the assessment of sustainability in the selected industry, and finally demonstrating the selected MCDM methods.

2.1. Sustainability in wood industries

Due to the presence of perceptible social and environmental challenges, sustainable development of supply chains has become a crucial issue in recent times. Many authors have discussed the different dimensions of sustainability in wood supply chain (Santos et al., 2019; Pérez et al., 2017). Among such research, Auer and Rauch (2021) discussed wood supply chain risks and risk mitigation strategies for effective wood supply chain operation. In terms of demand management, Hoogstra-Klein and Meijboom (2021) explored the possibilities and desirability of an increased demand orientation in wood supply chain. Life cycle assessment was considered by Peric et al. (2020) who conducted a study on wood supply

chain in the context of Serbia. However, these studies did not consider drivers and barriers in adopting sustainability in manufacturing sector. According to Santos et al. (2019), assessment studies needs to be considered in the evaluation of all dimensions of sustainability.

The notion of sustainability is brought up to nowadays life to protect the environment, improve the economy and provide healthy social life conditions (Fathollahi-Fard et al., 2021). The sustainability strategies embraced by firms should involve the three key constituents of sustainability: economic, social and environmental, which are together labeled by “triple base line” or 3BL (Burki et al., 2018). In the context of developed economy, Ouhimmou et al. (2019) proposed a generic decision support model to integrate the new product into value chain by considering sustainability in a forest value chain. In a related research, to evaluate the impact of economic sustainability of forest biomass, Solberg et al. (2020) introduced the concept of “wood for food”. This shows that evaluation, integration of different supply chain functions and assessment of sustainability of wood supply chain is not new for developed countries, however still lagging.

Wood is generally recognized as a recyclable resource of potentially high-volumes that are used in composite materials, hence, sustainability of this resource is a pillar for the survival of wood industry. The utilization of wood waste is experienced all over the globe by introducing new forms of wood products via series of transformation processes (Zezhou et al., 2019; Raphael et al., 2020). Generally, in wood manufacturing, only removal processes are applicable such as cutting, engraving, shaping, grinding, polishing which yield chips, dust, and fragments of trimmed wood pieces that can be recycled further by additional processing (Industries Summary, 2020).

2.2. Metrics and criteria of sustainability assessment

The articulation of environmental, social or economic sustainability requires performance indicators that are SMART, i.e., Specific, Measurable, Achievable, Realistic, and Timely (Selvik et al., 2021). The fundamental notion of sustainability development typically refers to achieving environmental cleanness, economic prosperity and communal responsibility, (Lerman et al., 2021). Quite a few scholars have explored how green SCM would safeguard the environment with an emphasis on green production, green content procurement, green processing, green marketing, green management of human capital, et cetera (Sunil and Subrahmanya Bhat, 2021). The current notion of SSCM concentrates equally on all three pillars of sustainability (Darbari et al., 2019; Allen et al., 2021). The presence of SSCM practices is gradually increasing in the published literature for the aim of achieving TBL (Tseng et al., 2020; Dyck and Manchanda, 2021). Previous literature studies show that many businesses are implementing sustainable manufacturing methods and environmentally managed systems to enhance the efficiency of the supply chain (Geissdoerfer et al., 2018); other researchers in short such as Simon (2019) presented a concept about the incorporation of circular economy activities in sustainable plastics manufacturing; Singh et al. (2020) established information and communication technology applications on food production for SSCM; Principato et al. (2019) explored the impact on sustainable development of circular economy applications in the Italian pasta industry; Xu et al. (2021) evaluated challenges to sustainable supply chains; Ponstein et al. (2019) examined sustainability in the production of Finnish wine; Reimann et al. (2019) established closed loop supply chains as a method for improvement in SSCM processes; Raut et al. (2019) attempted to relate the analytics of Big Data to organizational sustainability for sustainable business development; Gong et al. (2019) published a review on the role of consumer perception of sustainability; Gardas et al. (2019) have shown the critical success factors for reusability using waste to safe supply chains; Sharma et al. (2019) examined the sustainability effect of the circular economy on food supply chains; Saeed and Kersten (2019) attempted to categorize and recognize drivers to SSCM practices; Jabbour et al. (2019) demonstrated the effect on SSCM activities on the human side. All such recent studies emphasize on the importance of TBL concept in sustainability of industrial processes as well as circular economy which refers to a model of production and consumption, which involves leasing, sharing, repairing, reusing, refurbishing and recycling existing materials and products as long as possible. Table 1 shows a comparison between the selected articles in this literature brief in terms of the supply chain type, methodology; research type, etc.

3. Multi-Criteria Decision Making Techniques

Multi Criteria Decision Making is a branch of operations research that evaluates multiple conflicting criteria in decision-making problems (Dalalah et al., 2011). A. Decision-makers in industry, government, and medicine face conflicting criteria almost in every decision. The goal is to assist decision-makers in finding a compromised solution to problems of such a divisive nature. In most cases, there is no single best solution to this type of problems, so it is critical to use decision-makers' perspectives to distinguish between options. Many well established MCDM methods have been widely accepted in literature, among those we present the following:

- SAW—Simple additive weighting, which ranks the alternatives according to their weighted average.
- WPM—Weighted product method, the alternatives are compared by multiplying different ratios, raised to the power of the corresponding weight.
- AHP—Analytic hierarchy process, which is explained shortly in a subsequent section.
- VIKOR—“Vlsekriterijumska Optimizacija I Kompromisno Resenje” in Serbian, which translates to Multi-criteria optimization and compromise ranking, an approach that finds a compromise solution based on a stated ideal solution's proximity.

Table 1

Literature comparison by supply chain type, methodology, research type and focus.

Reference	Typical supply chain	Sustainable supply chain	Methodology	Type	TBL/Circular economy	Focus/Criteria
Biswajit et al. (2022)		✓	Two-stage supply chain		✓	Waste management
Santos et al. (2019)		✓	Optimization	Review		Wood supply chain
Carlos et al. (2022)				Review		Reuse of products
Pérez et al. (2017)		✓	Bibliographic analysis			Supply chain design
Dan et al. (2020)	✓		Inorganic analysis	Review		Reuse of products
Auer and Rauch (2021)	✓		Risk mitigation	Review		Risk
Saeed et al. (2022)		✓	Closed loop supply chain	Optimization		Recyclable wastes
Berger et al. (2020)	✓		Building Material	Recycling		Recyclable wastes
Reimann et al. (2019)	✓		Closed loop	Process innovation		Remanufacturing
Farah et al. (2022)			Insulation analysis	Energy savings		Environmentally friendly materials
Suhi et al. (2019)		✓	Assessment		✓	Environmental Impact
Hoogstra-Klein and Meijboom (2021)		✓	Qualitative analysis	Demand Management		Wood Products
Mohammadi (2020)	✓		Lot sizing	Demand Uncertainty		Hazardous materials
Zezhou et al. (2019)	✓		off-site snapshot methodology	Case study		Waste composition
Riccardo et al. (2022)		✓	Simulation			Packaging
Ouhimmou et al. (2019)		✓	Assessment			Decision Support Systems
Saavedra et al. (2018)		✓	Systems dynamic approach			Renewable resources
Raphael et al. (2020)			comparative evaluation	Case study		Environmental Impact
Simon (2019)	✓		Overview	Review		Plastics
El-Amrani et al. (2021)		✓	Bayesian Network	Modeling and Assessment		Multi Echelon
Gardas et al. (2019)		✓	Quantitative approach	Critical success factors		Plastic Packaging
Tseng et al. (2020)		✓	Bibliometric analysis		✓	Triple bottom line
Anselmo and Piedra-Muñoz (2021)			Life cycle	Case study		Non-renewable resources

(continued on next page)

- TOPSIS—Technique for order preference by similarity to ideal solution, which identifies ideal and negative-ideal solutions and all other alternatives are ranked based on their distance from the ideal and negative ideal solutions.
- ELECTRE III—Elimination and choice translating reality, which is based on an outranking relation between the alternatives.
- BWM—Best-Worst Method, is explained shortly in a subsequent section.
- Decision trees—An approach that identifies possible payoffs and their associate probabilities to estimate the choice with the highest expected payoff.
- ANP—Analytic Network Process is a generic variant of the AHP that may overcome hierarchical structural constraints and is suggested for decision-making situations involving cross relations in outer dimensions.
- DEA—Data Envelopment Analysis, an optimization technique to find the efficacies of decision making units, however, it also can help identify the best alternative decision making unit.
- Fuzzy MCDM—here, fuzzy domain is applied in which variable are expressed in terms of linguistic variables that have a degree of overlap. Fuzzy MCDM can be incorporated in many MCDM methods.

In this research, three decision-making tools are going to be used: Hierarchical Clustering, Analytical Hierarchy Process and Best-Worst Method. A brief literature about MCDM and its applications is presented in the next section.

3.0.1. Analytical Hierarchy Process

When decision-makers encounter a mix of qualitative, quantitative, and sometimes conflicting criteria, robust tools should be used to shape coherent decisions. Among these tools, AHP is found to be efficient in optimizing the decision-making process. Developed in the 1970s by T. L. Saaty, AHP is still one of the most common decision support tools to date

Table 1 (continued).

Reference	Typical supply chain	Sustainable supply chain	Methodology	Type	TBL/Circular economy	Focus/Criteria
Sakib et al. (2021)	✓		Bayesian belief networks			Oil and gas
Xu et al. (2021)	✓		Comparison approach	Case study		Forests Policies
Jian-Xiu and Zhao (2020)			Economic Impact analysis			ISO 14001 certification
Darbari et al. (2019)			Fuzzy criteria programming approach		✓	Closed loop supply chain
Khan and Wuest (2018)	✓		Framework			Products and services
Fathollahi-Fard et al. (2021)		✓	Flow shop scheduling	Permutation	✓	Triple bottom line
Dogaru (2020)		✓	Sustainable Development analysis			Environmental budget
Burki et al. (2018)	✓		Manufacturing concepts	Transportation	✓	triple bottom line
Shigui et al. (2021)		✓	Technology Investment analysis			Government Regulations
Axon (2020)		✓	Soco cultural analysis	Transformational change		Workers
Peric et al. (2020)	✓		Life cycle assessment	Case study		Wood chips
Solberg et al. (2020)			Economic impacts analysis	Policies		Food
Saeed and Kersten (2019)		✓	Classification			Drivers to sustainability
Selvik et al. (2021)			SMART analysis	Performance indicators		Quality
Lerman et al. (2021)			TBL analysis	Renewable energy systems	✓	A triple bottom line
Raut et al. (2019)	✓		Big Data analysis			Sustainable businesses
Dalalah (2016)	✓		Simulation–optimization	Case study		Discrete events
Zamiela et al. (2021)	✓		PIV) MCDM			Healthcare
Sunil and Subrahmanya Bhat (2021)			Supply chain practices	Review		Materials
Allen et al. (2021)		✓	Circular economy management			circular economy
Geissdoerfer et al. (2018)	✓		Circular Approach	Strategic Planning	✓	Business models
Singh et al. (2020)	✓		ICT	Benchmarking		Organizational performance
Principato et al. (2019)			Quantitative approach	Case study	✓	Food waste and loss
Ponstein et al. (2019)	✓		Emission Analysis			Greenhouse emissions
Gong et al. (2019)		✓	Fixed-effect model	Practice		Customer awareness
Sharma et al. (2019)		✓	Circular Economy approach			Food supply chain
Dyck and Manchanda (2021)			Virtue ethics Analysis	Marketing concepts		Socio-ecological
Jabbour et al. (2019)	✓		Circular approach	Review	✓	Human

that is capable of synthesizing intricate decisions. A typical decision problem should consist of alternatives and conflicting criteria to fit the framework of AHP. Thoughtful decision-making includes weighing the alternatives and all the factors that characterize the decision problem. While, there are many well established approaches used in the decision-making process under the condition of multi-criteria such as TOPSIS, AHP, VIKERS, EVAMIX and PROMETHEE, AHP is dominating modern scientific research (Yagmur, 2016; Tang et al., 2014). It has been used in many applications as those listed in (Lyu et al., 2020; Abhishek and Kumar, 2018). Specifically, several application of AHP can be found in literature such as reverse logistics service provider selection (Jain and Khan, 2017); Crane selection in construction sites (Dalalah et al., 2010); hospital waste management (Ishtiaq et al., 2018) and ranking of univariate forecasting techniques (Naim et al., 2020) are few to name. It has been used successfully in several contexts such as health, environment, social studies and technology (Jovanović et al., 2015; Emrouznejad et al. 2017). AHP has also been used with integration of different MCDM methods such as fuzzy methods (Khan et al., 2019). The Hierarchy process approach delivers an informative and effectual calculation of the comparative weights of several factors for complex assessment problems. It has the advantage of allowing the criteria to have a hierarchical structure. It also let users shed the light on specific criteria and their descending sub-criteria in a decision tree (Sui et al., 2016).

In this research, after conducting the hierarchical clustering, AHP will be used for further assessment and ranking as will be illustrated shortly. The reason for selecting both AHP and the Best-Worst method is that, AHP is a widely used MCDM method since its development. Its applications can be found in service sector, manufacturing sector, pharmaceutical sector, automotive sector and many others. Several authors over the last few decades used AHP in prioritizing criteria and its associated sub-criteria due to many reasons. First, AHP divides the complex problem into simpler hierarchy to accommodate consistency requirements in the decision makers' judgments, secondly, it allows less number of pair-wise comparison as compared to other MCDM methods, and lastly, it removes biasness in expert's judgment.

3.0.2. Best-Worst-Method (BWM)

Developed by Rezaei (2015), the best-worst method is a relatively new multi-criteria decision-making approach that is characterized by distinct features. It determines the best weights of a group of criteria based on pairwise preferences of a single decision-maker (Mohammadi and Rezaei, 2019). While AHP is one of the most common comparison-based multi criteria decision methods which involves a pairwise judgment of all " n " decision criteria, i.e., $n(n-1)/2$ pair-by-pair judgments, the best-worst method only needs reference-pairwise comparisons, yielding " $2n-3$ " comparisons, yet still maintains the integrity of the decision problem. Anyway, almost all MCDM approaches are designed for individual assessment. In the so-called group decision-making, such MCDM approaches cannot integrate the preferences of multiple evaluators without further preparation. Rezaei (2015) proposed that the best-worst reference techniques can be used to efficiently and reliably address multi-criteria problems with less needs for comparative data; thus, it does not necessitate full pairwise judgment matrices and is therefore less complex, time-saving and easily implemented compared to other MCDM methods.

Pursuant to Rezaei et al. (2016), for a completely consistent comparison that involves few parameters, BWM may generate a single solution. BWM may find multiple optimal solutions for inconsistent system. It is evident from the emerging literature that the BWM is gaining more popularity and has begun to appear in different industrial applications taking into account its special features. BWM has been related as an effective MCDM technique in assessing the environmental performance of supply chains. In spite of its potential to solve intricate decision-making problems, it was not widely implemented in industrial settings. This study adds to the existing literature by the use of Hierarchical Clustering, AHP and BWM to weigh environmental, social and economic factors. Among the reasons for selecting BWM, in short, we list the following:

- BWM demands less pairwise data while allowing for more consistent comparisons.
- Pairwise comparisons have the further benefit of allowing to quantify the inconsistency of the decision makers' preferences.
- BWM is compatible with other MCDM approaches and can be integrated seamlessly with other methods.
- It is well-suited for group decision-making and leads to more trustworthy weights/rankings.
- It employs optimization to determine the weights while other methods require prior quantification of such weights.
- It uses the de-biasing approach by considering "the-opposite" to draw conclusions in an organized fashion.

3.1. Criteria for sustainable wood supply chain

A key concern for the forest industry in these times of increased competition is how to produce wood in an efficient and cost-effective manner while also completely meeting consumer demands (Marjanke et al., 2021). One idea is to strengthen the wood supply chain's management as well as its planning and coordination procedures. This can be achieved by identifying the criteria that is employed to assess the operation of sustainable supply chains. One of the main obstacles to assessing environmental efficiency is the intrinsic complexity of its multi-criteria. With the help of existing literature and a leading furniture manufacturer in the Middle East (Woodpecker L.L.C), the authors could gather information from different Woodpecker partners in the region. The data regarding the key indicators of environmental sustainability in Table 2 is established based on the findings of our literature review and the gathered information from different branches of Woodpecker.

4. Methodology

According to Table 2, environmental sustainability in wood supply chain is a multi-criteria concept, thus in order to assess the relevance of the various criteria; decision making tools may be helpful. In this study, we use the novel triplet of HC-AHP-BWM techniques as a research methodology, which to the best of our knowledge, has not been used in this area before. To eliminate the number of collected criteria, first, a hierarchical clustering (HC) is performed to reduce the data and remove any redundancy. After the criteria are clustered into smaller groups, an AHP-BWM analysis will be conducted and recommendations will be suggested accordingly as shown in Fig. 1.

Table 2

Environmental sustainability indicators in wood supply chain according to our surveys and literature studies.

	Criteria	Brief description	References
1.	Waste management	This criterion describes generally how an industry deals with the wastes produced (recycle, or disposal). Vast quantities of wastes are produced every day (High solid, low liquid, or moderate gaseous, chemical waste) which have a negative impact on the environment causing air and land contamination by toxic wastes, and possible loss of fertility amongst the local inhabitants. Wood industries result in wastes such as timber, sawdust, shavings, boards trims/veneer panel, and bonding chemicals.	Biswajit et al. (2022) , Acquaye et al. (2017) , Dong and Hauschild (2017) and Schiefer et al. (2015, 2016)
2.	Reuse of products	Involves the use of the material of the discarded products to help reduce waste by remanufacturing and recycling. In our context, wood waste or used wood products can be utilized as raw materials for other products such as plywood, medium density fiberboard (MDF) and wood treated with Copper Chrome Arsenate (CCA).	Carlos et al. (2022) , Golpîra et al. (2017) , Kremer et al. (2016) and Dan et al. (2020) .
3.	Recyclable wastes	Recycle excess materials, wood chips, dust, and debris by contracting with disposal and recycle companies that take care of the residues.	Saeed et al. (2022) , Faraca et al. (2019) , Berger et al. (2020) and Shyusuke et al. (2019) .
4.	Environmentally friendly materials	Involves the usage of products that do not harm the environment whether in manufacturing or disposal such as Bamboo and friendly chemicals used to bond MDF, CCA, particle board wood and paints.	Farah et al. (2022) , Kremer et al. (2016) and Dong and Hauschild (2017) .
5.	Hazardous materials	Involves the usage of toxic materials such as ethanol, acetone, lead, arsenic, Chlorofluorocarbons emissions and other harmful chemicals that can cause damage to natural ecosystems, the environment and human health.	Mohammadi (2020) , Wan Ahmad et al. (2017) , Ding et al. (2015) , Aljazzar et al. (2018) , Wan Omar et al. (2014) , Viglia et al. (2016) , and Song et al. (2017) .
6.	Packaging	Involves the usage of non-biodegradable materials in wood packaging and the use of available resources (e.g., raw material, fuel and space) appropriately.	Riccardo et al. (2022) , Chen and Chen (2017) , Pimenta and Ball (2015) .
7.	Renewable resources	It involves the usage of different types of renewable energy sources such as solar, hydropower, geothermal, biomass, and others which could somehow reduce pollution. In wood industry context, it refers to the consumption of wood from forestry and pollution by liquid and solid waste which contaminate lakes, rivers, oceans that promote the aquatic wildlife.	Mohammad Ruhul et al. (2021) , Saavedra et al. (2018) , Viglia et al. (2016) , Aivazidou (2016) and Acquaye et al. (2017) .
8.	Non-renewable resources	The normal energy sources, including oil, coal, and natural gas, contain residuals. Overuse of these energy sources can harm the atmosphere and trigger health problems. Carbon material dissipation is a major cause of global warming.	Anselmo and Piedra-Muñoz (2021) , Kremer et al. (2016) , Dong and Hauschild (2017) , Chen and Chen (2017) , and Blanco et al. (2016) .
9.	ISO 14001 certification	A multinational environmental standards which help organizations reduce their operations that have impact on the environment via customary requirements and environmental legislation.	Jian-Xiu and Zhao (2020) , Ferreira et al. (2016) , Arimura et al. (2016) and Pimenta and Ball (2015) .
10.	Products and services	Involves the products and services that are provided by the wood industry whether they are environmentally friendly or not.	Khan and Wuest (2018) and Aihua et al. (2017) .
11.	Environmental budget	Involves the financial amounts that the company invests in environmental sustainability.	Dogaru (2020) and Wen-Hui et al. (2022) .
12.	Government Regulations	Involves regulations provided by the government that require industries to follow legislative policies in order to protect the environment.	Shigui et al. (2021) , Ding et al. (2016) and Wan Ahmad et al. (2017)
13.	Workers implication in environmental sustainability	Involves workers and employees in activities performed in order to improve environmental sustainability.	Axon (2020) and Clarke-Sather and Cobb (2019) .

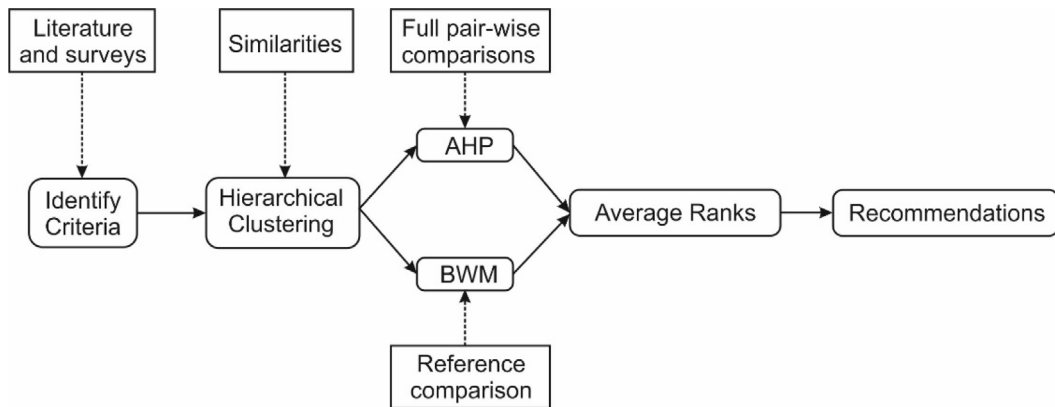


Fig. 1. Study methodology with dotted lines as inputs.

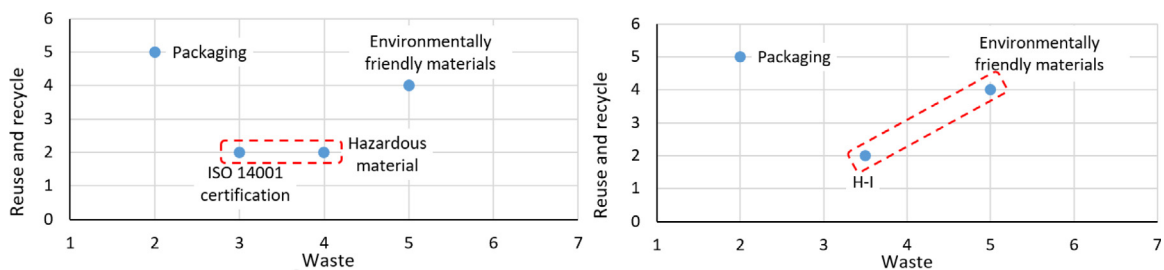


Fig. 2. Clustering of four factors.

4.1. Clustering technique

Hierarchical Clustering (HC) is somehow a data reduction technique which involves creating ordered clusters from top-down/bottom-up of objects according to their similarity with respect to a set of criteria. HC has been used in various research areas to cluster objects according to similarity into hierarchical groups (Popescu Gheorghe et al., 2017; Chang et al., 2021). Hierarchical clusters are more informative than other clustering types due to the resulting Dendrogram structure that shows the way objects are allocated to their parent clusters and the hierarchical relationship between objects either from top-down (divisive approach) or bottom-up (agglomerative approach).

One advantage of HC is that it does not require to pre-specify the number of clusters. Agglomerative clustering is used in this study since it works in a bottom-up manner. That is, each object is initially considered as a single cluster. Then the most similar clusters (i.e., the least distant) are combined into a new bigger cluster. This procedure is iterated until all resulting clusters are members of one single big cluster. The distance between clusters can be found using different link types. The average link approach is used in this study to calculate the distance between the clusters, which simply takes the average Euclidean distance between all the objects of any two clusters. To illustrate, we purposely select four factors from Table 2 with respect to two criteria, namely, Waste and Reuse. Fig. 2 depicts the location (i.e., the value) of each factor. Using the Euclidean distance, one can see that ISO 14001 and hazardous material form the closest pair, where the distance between an observation $(X_{1i}, X_{1i}, \dots, X_{ki})$ and another $(X_{1j}, X_{1j}, \dots, X_{kj})$ is given by:

$$D(i, j) = \sqrt{(X_{1i} - X_{1j})^2 + \dots + (X_{ki} - X_{kj})^2}$$

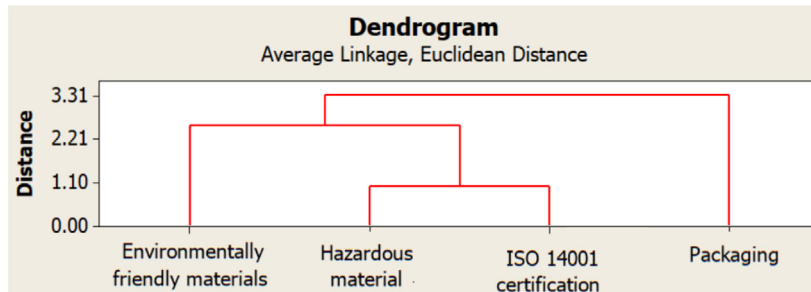
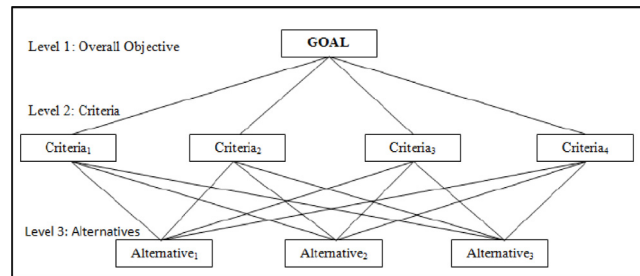
where observation i and j are distinct by k dimensions. The resulting distances are given in Table 3, where each factor is denoted by its initial letter. With this, H and I will make the first cluster (H-I). The next iteration is shown in middle part of Table 3, where the distance from the cluster H-I to any other factor is given by the average of all its constituents to the associated factor. For instance, the distance from cluster H-I to E is computed as $(2.24 + 2.83)/2 = 2.53$. Iteration 2 shows that H-I and E will make the new cluster. Finally, the last factor will join the last layer of clusters, thereby, making 3 clusters as shown in Fig. 2 and the Dendrogram of Fig. 3.

As a remark, Fig. 3 does not necessarily show the final clustering in this studies, since more dimensions are to be considered as will be illustrated shortly.

Table 3

Distances between clusters, the factors are marked by their initial letters and shortest pair by bold type.

	Iteration 1				Iteration 2				Iteration 3		
	E	H	P	I	E	H-I	P		H-I-E	P	
E	0	2.24	3.16	2.83	E	0	2.53	3.61	H-I-E	0	3.31
H		0	3.61	1	H-I		0	3.38	P		0
P			0	3.16	P			0			
I				0							

**Fig. 3.** Hierarchical clustering illustrated.**Fig. 4.** AHP Hierarchy.

4.2. Analytical hierarchy process

The typical steps of AHP are describe as follows:

Step 1: Define problem/set objective and list of criteria that are essential for achieving objectives or related to defined problem.

Step 2: Structure the problem into different hierarchical levels. First hierarchy defines the objectives, second level hierarchy defines the set of major criteria, third level hierarchy defines the sub-criteria associated with each main criteria and the fourth hierarchy defines the available alternatives. Fig. 4 shows the general structure of AHP hierarchy.

Step 3: The decision maker performs a pairwise comparison based on a scale (from 1 to 9) between major criteria and sub-criteria with respect to major criterion. If “ n ” criteria are considered in the model, the number of pairwise comparisons will be $\frac{n^2-n}{2}$. The matrix shown in (1) embodies the pairwise comparisons.

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad (1)$$

Step 4: After pair-wise comparison, in this step, the relative weights of the criteria are calculated. Such computation involves the estimation of the normalized Eigenvector from the specified matrix A . The Eigenvector (w) corresponding to the largest eigenvalue λ_{max} provides the relative weights as mentioned below.

$$Aw = \lambda_{max}w \quad (2)$$

Table 4

The standard AHP random index.

Size of comparison matrix	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49

Step 5: In order to check the consistency of experts' judgments, the consistency must be checked for each comparison matrix at each stage. A consistency index (*CI*) is determined to test the accuracy of the comparison matrix. Below equation shows the formula used to find the consistency index.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (3)$$

The consistency ratio (*CR*) is finally calculated. The *CR* checks whether the assessments are adequately consistent. The *CR* can be determined by taking the ratio between *CI* and the Random Index (*RI*):

$$CR = \frac{CI}{RI} \quad (4)$$

where the *RI* is given in Table 4.

The acceptance bound for the *CR* is 0.1. If the *CR* value exceeds 0.1, it is recommended that a pair-wise analysis be made again in order to render the judgment more clearly.

Step 5: Find the overall criteria weights by conducting matrix algebra for all the evaluations in the hierarchy.

4.3. Best Worst Method

The BWM is based on a systematic comparison of the decision criteria. The approach is used to test a collection of alternatives against a set of parameters for comparison purposes. This is, once the decision-maker (DM) has defined the judgment criteria, the DM chooses two parameters: the best criterion, and the worst criterion. The best criterion is the one that plays a positive role, whereas the worst criterion plays the opposite role with respect to the main goal. Using a predefined scale (e.g. 1 to 9), the DM then gives the preference of the best criterion over all other criteria, and then over the worst criterion. The BWM is demonstrated in the following steps:

Step 1:

Define the group of decision criteria. The group of decision criteria are denoted by $C = \{c_1, c_2, \dots, c_n\}$.

Step 2:

The decision maker specifies the best decision criterion (*B*) which is the most significant and the worst decision criterion (*W*) which is the least important or the least desirable from the set *C* where $B, W \in C$.

Step 3:

In this step, a pairwise comparison between the best (*B*) and the other criteria from *C* is performed by the decision maker, such that integers from 1 to 9 are used, where 1 means equally important and 9 means extremely more important. The resultant vector is called best-to-others, which is given by:

$$A_B = (a_{B1}, a_{B2}, \dots, a_{Bn}),$$

where, a_{Bj} expresses the significance of the most important criterion *B* over criterion *j*, and thus $a_{BB} = 1$ as shown in Fig. 5.

Step 4:

The decision-maker compares the worst (*W*) criterion with the other criteria of *C* in a pairwise manner, where the same 1–9 scale is used. This step will generate worst-to-others vector which is given by:

$$A_W = (a_{1W}, a_{2W}, \dots, a_{nW})^T,$$

where a_{jW} is the evaluation of criterion *j* over the worst criterion *W* and $a_{WW} = 1$.

Step 5:

In this step, the optimal weights $w^* = (w_1^*, w_2^*, \dots, w_n^*)$ are found. Of note, for a fully consistent system, the absolute differences $\left| \frac{w_B}{w_j} - a_{Bj} \right|$ and $\left| \frac{w_j}{w_W} - a_{jW} \right|$ for $j = 1, 2, \dots, n$ should be zeros. Therefore, given A_B and A_W , a weight vector w^* can be computed such that these differences are minimized. The weight vector must be in the vicinity of the terms $w_B/w_j = a_{Bj}$, and $w_j/w_W = a_{jW}$ for $j = 1, 2, \dots, n$. Thus, the maximum absolute differences $\left| \frac{w_B}{w_j} - a_{Bj} \right|$ and $\left| \frac{w_j}{w_W} - a_{jW} \right|$ for $j = 1, 2, \dots, n$ can be minimized as follows:

$$\min_w \max_j \left\{ \left| \frac{w_B}{w_j} - a_{Bj} \right|, \left| \frac{w_j}{w_W} - a_{jW} \right| \right\} \quad (5)$$

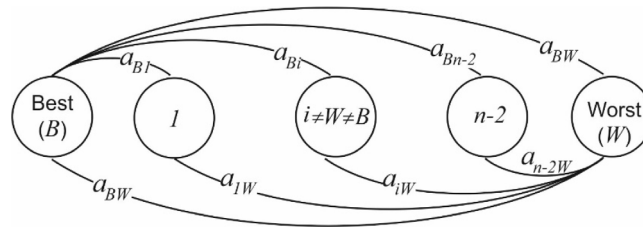


Fig. 5. The best and worst pairwise comparison with others.

s.t.

$$\sum_{j=1}^n w_j = 1$$

$$w_j \geq 0 \quad \forall j$$

By manipulating the above primal problem, a simpler form will result as follows:

$$\begin{aligned} \min \xi & \\ |w_B - w_j a_{Bj}| &\leq \xi \quad \forall j \\ |w_j - w_W a_{jW}| &\leq \xi \quad \forall j \\ \sum_j w_j &= 1 \\ w_j &\geq 0 \quad \forall j \end{aligned} \tag{6}$$

where the value of ξ indicates the consistency level of the decision makers' opinions. By solving the above model, the weights $\mathbf{w}^* = (w_1^*, w_2^*, \dots, w_n^*)$ and the consistency ratio ξ can be determined. When the value of ξ is close to zero, a high consistency level is observed, while larger values indicate the opposite and the evaluations have to be revised in this case, (Rezaei, 2016). Of course, a value of "0" indicates a full consistency. The model above can be solved using the simplex method or any standard linear optimization solver.

5. Real world application

5.1. Case company description

Woodpecker Company is a leading manufacturer of furniture in the Middle East, with its headquarters located in UAE. It was established in 1992, with all its branches in the Middle East equipped with the latest, state-of-art machineries and tools which are managed by a team of professionals, designers, skilled carpenters and painters, and other craftsmen. The company has a certified manufacturing units that provide quality solutions at reasonable costs, enabling them to pass on handsome savings to their customers. Being an entrepreneur in furniture making, Woodpecker supplies a variety of products to all Emirates in UAE and Gulf region. With a record of accomplishment of many years, their skillful and experienced workforce ensures successful and timely deliveries. Woodpecker has been heavily involved in the data collection of this study. The data collected in this study was gathered from different branches of Woodpecker Company in the Middle East and Gulf region during the academic year of 2019–2021.

5.2. Survey design and data collection

Based on the exploration of the existing literature and surveys, the criteria of our decision problem could be identified as outlined in Table 1. A questionnaire-based survey was delivered to a group of 20 experts/academicians to pick and rate the most relevant indicators, among which 17 respondents did provide reasonable evaluations. Most of the industrial managers and academic specialists have over 10 years of experience in his field. The academicians were drawn from selected institutions who had industrial development expertise. Some demographic data of the respondents is shown in Table 5.

In the following sections, clustering of the criteria followed by AHP–BWM framework will be demonstrated using the collected data and ultimately, by measuring the relative weight of each criterion we will be able to identify the most important factors.

Table 5
Demographic data of Experts/Academicians.

Experience	N = 17	%
3 to 5 years	2	12%
6 to 10 years	3	18%
11 to 15 years	12	71%
16 years +	0	0%
Education		
PhD	2	12%
MS	4	24%
MBA	0	0%
BE	11	65%
Designations		
Deputy Manager	5	29%
Manager	4	24%
Senior Manager	4	24%
Deputy General Manager		0%
General Manager	4	24%
Department		
Supply chain	8	47%
Production	4	24%
Planning	4	24%
Health Safety Eng.	1	6%

Table 6
The final approved distance measures between the criteria and their clusters.

Clustering weights						
	Waste	Reuse and recycle	Materials	Resources	Environmental certification	Others
Waste management	10	5	2	1	4	2
Recyclable wastes	6	10	4	2	1	1
Reuse of products	3	10	3	1	3	1
Environmentally friendly materials	5	4	10	3	4	2
Hazardous material	4	2	10	2	1	0.5
Packaging	2	5	10	2	2	1
Renewable resources	2	2	1	10	4	4
Non-renewable resources	2	2	1	10	4	4
ISO 14001 certification	3	2	5	2	10	2
Products and services	4	1	3	1	5	10
Environmental budget	3	5	2	1	6	10
Government regulations	3	5	2	3	1	10
Workers implication in environmental sustainability	5	4	3	1	1	10

5.3. Cluster analysis of the criteria

The relationship between the set of criteria is collected from the assigned experts. After multiple rounds between the experts, the approved values are shown in Table 6. The average distances were rounded to the nearest integer. For instance, the experts' assessment shows that the criterion "Renewable Resources" is 2 to 10 in its closeness measure to "Waste Management". Thirteen criteria were selected which were grouped using Hierarchical Clustering based on their similarities given in Table 6. Minitab software is used to calculate the distances and obtain the final clusters as shown in the Dendrogram of Fig. 6. Any group of criteria having a similarity level of 85% or more were considered as one cluster. Hence, a set of criteria that has less similarity than this threshold will be split into additional clusters. The analysis shows that 6 distinct clusters could be found as shown in Fig. 6.

5.4. Application of Analytical Hierarchy Process

After clustering the significant criteria using the Dendrogram analysis, the first step in AHP is to perform a pairwise comparison between the selected criteria clusters using the standard AHP scale, where "1" refers to Equally preferred pair, "3": Moderately preferred, "5": Strongly preferred, "7": Very strongly preferred, "9": Extremely preferred and 1, 4, 6, 8 are intermediate values between the direct judgments. Table 7 shows the final round of pairwise evaluations of the 17 experts for the six resulting criteria.

Table 8 depicts the pairwise evaluations of the sub-criteria in regard to "Reuse and Recycle". Since only three sub-criteria are included, one single pairwise comparison is needed. The experts showed that recycling is as important as reuse with a rating of "equally preferred", i.e., 1.

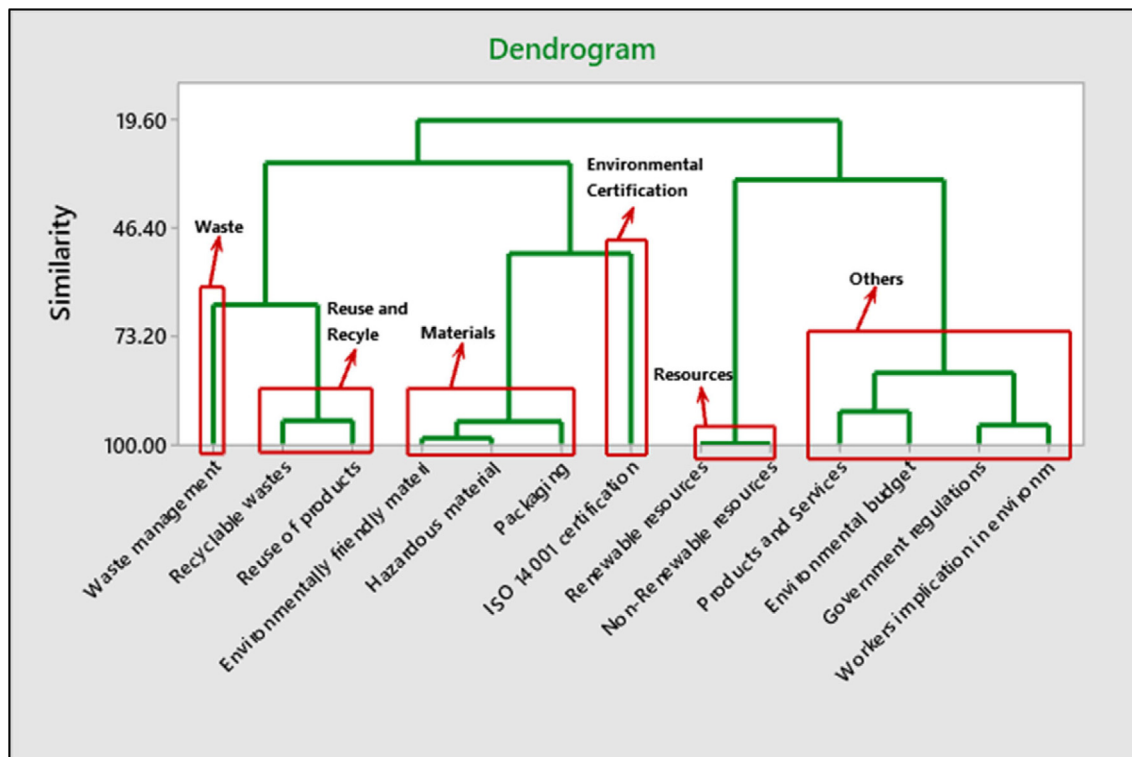


Fig. 6. Resulting Dendrogram with adjacent sub-criteria clustered together.

Table 7

Final round of pairwise comparison of the top level of the hierarchy.

Pairwise comparison	Waste management	Reuse and recycle	Materials	Resources	Environmental certification	Others
Waste management	1	2	3	5	7	9
Reuse and recycle	1/2	1	3	4	6	9
Materials	1/3	1/3	1	2	4	6
Resources	1/5	1/4	1/2	1	2	3
Environmental certification	1/7	1/6	1/4	1/2	1	2
Others	1/9	1/9	1/6	1/3	1/2	1
Incon:0.02						

Table 8

Pairwise comparison with respect to Reuse and Recycle.

Pairwise comparison	Reuse of products	Recyclable wastes
Reuse of products	1	1
Recyclable wastes	Incon:0.00	1

Table 9

Pairwise comparison with respect to Materials.

Pairwise comparison	Environmentally friendly materials	Hazardous material	Packaging
Environmentally friendly materials	1	1/7	2
Hazardous material	7	1	6
Packaging	1/2	1/6	1
Incon:0.04			

The pairwise comparison of the sub-criteria relating to “Material” shows that hazardous material has more impact with respect to the produced or used material as shown in Table 9. The pairwise evaluation of the sub-criteria with respect to “Resources” shows that renewal resource is 10 times as important as the non-renewable resources as shown in Table 10.

Finally, the pairwise comparison of the “Others” category is given in Table 11, where budget seems to be significantly important when compared to workers implications. Fig. 7 shows the weights of all sub-criteria

Table 10

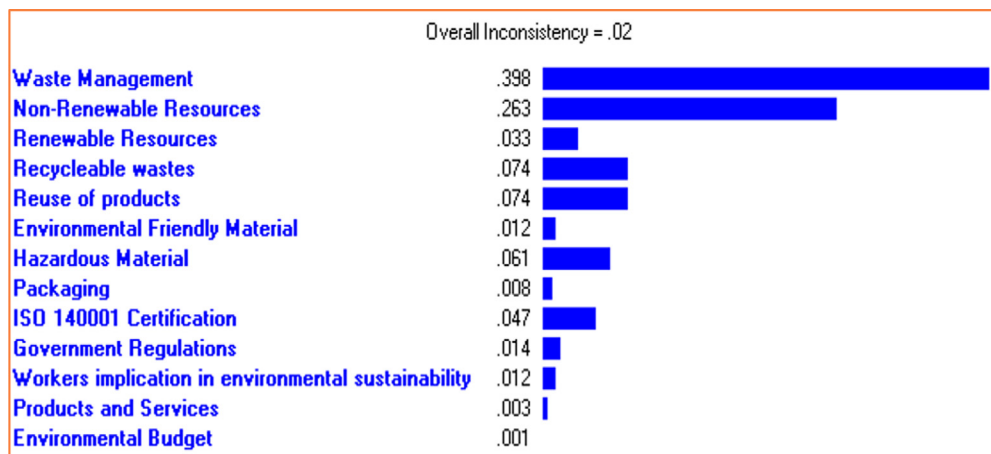
Pairwise comparison with respect to Resources.

Pairwise comparison	Renewable resources	Non-renewable resources
Renewable resources	1	10
Non-renewable resources	1/10	1
Incon:0.00		

Table 11

Pairwise comparison with respect to "Others".

Pairwise comparison	Products and Services	Environmental budget	Government regulations	Workers implication in environmental Sustainability.
Products and services	1	3	1/4	5
Environmental budget	1/3	1	1/6	1/9
Government regulations	4	6	1	2
Workers implication in environmental sustainability	1/5	9	1/2	1
Incon:0.08				

**Fig. 7.** Weights of all sub-criteria.

From the collected data and the above pairwise comparison, we proceed in AHP and synthesize them with respect to our goal, that is, Environmental Manufacturing Sustainability as shown in Fig. 8. The overall consistency is 0.08 which is acceptable, i.e., ≥ 0.1 .

Fig. 8 above demonstrates the optimal weights of the manufacturing environmental sustainability criteria produced by AHP. The criterion with the highest weight is said to have the highest impact, whereas the criterion with the lowest weight is said to have the lowest impact in evaluating manufacturing environmental sustainability. In this study, "Waste Management" is the criterion with the highest weight of 0.398 while "Others" is the criterion with the lowest weight of 0.030. Furthermore, the consistency ratio obtained equals 0.08, which is acceptable since it is less than 0.1, and that indicates that the evaluations are appropriately consistent. The same criteria are also evaluated using the BWM. The next section demonstrates the BWM analysis.

5.5. Application of Best Worst Method

The BWM starts by identifying the best and worst criteria to be compared when compared with the remaining criteria. Each survey respondents was asked to identify the criteria they believe it is the most important (best) and the least important (worst) in the setting of the designated list of indicators. Similar to those used in AHP, a rating from 1–9 is usually used for the pairwise comparison. For instance, expert number 3 identified "Resources" as the best and "Environmental certification" as the least important. His evaluation of the best-to-others is shown in Table 12 and worst-to-others is shown in Table 13.

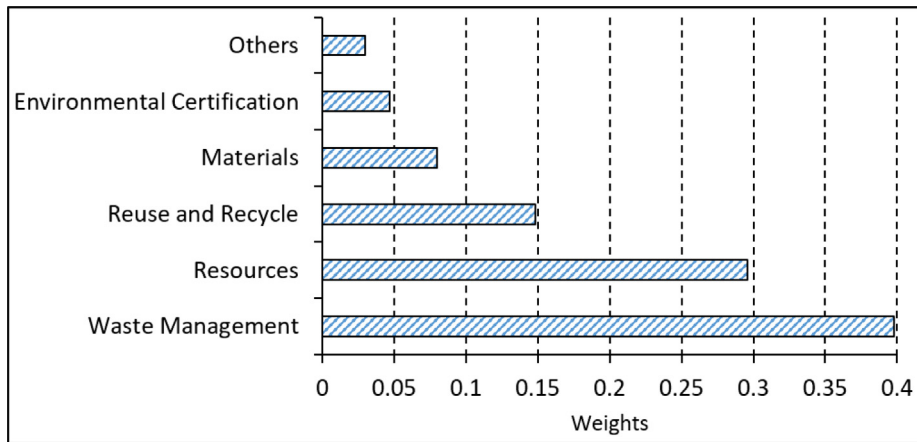


Fig. 8. Optimal weights obtained from AHP with an inconsistency ratio of 0.08.

Table 12

Vector of Best-to-others .

	c_1	c_2	c_3	c_4	c_5	c_6
Best to others	Waste management	Reuse and recycle	Material	Resources	Environmental certification	Others
Resources	2	5	2	1	9	2

Table 13

Vector of Others-to-worst.

	c_1	c_2	c_3	c_4	c_5	c_6
Others to worst	Waste management	Reuse and recycle	Materials	Resources	Environmental certification	Others
Environmental certification	7	7	5	9	1	7

Table 14

Optimal weights obtained from BWM.

Criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Ave. weights
Waste management	0.412	0.331	0.197	0.319	0.377	0.299	0.364	0.377	0.377	0.300	0.431	0.324	0.362	0.414	0.321	0.330	0.329	0.345
Resources	0.196	0.233	0.305	0.293	0.255	0.253	0.213	0.178	0.252	0.297	0.211	0.223	0.313	0.239	0.251	0.315	0.342	0.259
Others	0.102	0.143	0.197	0.133	0.115	0.176	0.144	0.165	0.117	0.172	0.111	0.124	0.089	0.141	0.129	0.080	0.080	0.131
Materials	0.137	0.150	0.197	0.094	0.085	0.134	0.146	0.152	0.084	0.097	0.106	0.143	0.102	0.099	0.144	0.124	0.124	0.125
Reuse and recycle	0.116	0.122	0.079	0.116	0.081	0.104	0.106	0.091	0.141	0.102	0.110	0.150	0.108	0.087	0.122	0.119	0.119	0.110
Environmental certification	0.038	0.021	0.024	0.022	0.090	0.033	0.028	0.037	0.028	0.031	0.032	0.037	0.026	0.021	0.032	0.035	0.015	0.031
Consistency ratio	0.072	0.079	0.089	0.109	0.098	0.103	0.067	0.076	0.052	0.079	0.106	0.089	0.081	0.055	0.108	0.132	0.158	0.093

In BWM, an optimization model will result for each expert, for example, the evaluations of expert #3 will result in the following model:

$$\begin{aligned}
 & \min \xi \\
 & |w_B - 2w_1| \leq \xi; |w_1 - 7w_W| \leq \xi \\
 & |w_B - 5w_2| \leq \xi; |w_2 - 7w_W| \leq \xi \\
 & |w_B - 2w_3| \leq \xi; |w_3 - 5w_W| \leq \xi \\
 & |w_B - 1w_4| \leq \xi; |w_4 - 9w_W| \leq \xi \\
 & |w_B - 9w_5| \leq \xi; |w_5 - 1w_W| \leq \xi \\
 & |w_B - 2w_6| \leq \xi; |w_6 - 7w_W| \leq \xi \\
 & w_1 + w_2 + w_3 + w_4 + w_5 + w_6 = 1 \\
 & w_j \geq 0 \quad \text{for all } j = 1, \dots, 6
 \end{aligned} \tag{7}$$

where for expert #3 the most important criterion is c_4 and the least one is c_5 , hence, $w_B = w_4$ and $w_W = w_5$. Fig. 9 shows the modified source file of the excel solver that finds the optimal solution of this problem instance. The optimal weights for expert #3 are shown in Fig. 10.

A similar analysis is conducted for the remaining experts. Table 14 shows all the experts weights and the resulting arithmetic average. The consistency ratio of all experts is found to be acceptable with an average value of 0.093.

Criteria Number = 6	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Criterion 6
Names of Criteria	Waste Management	Reuse and Recycle	Materials	Resources	Environmental Certification	Others

Select the Best	Resources
Select the Worst	Environmental Certification

Best to Others	Waste Management	Reuse and Recycle	Materials	Resources	Environmental Certification	Others
Resources	2	5	2	1	9	2

Others to the Worst	Environmental Certification
Waste Management	7
Reuse and Recycle	7
Materials	5
Resources	9
Environmental	1
Others	7

Weights	Waste Management	Reuse and Recycle	Materials	Resources	Environmental Certification	Others
	0.197255575	0.07890223	0.19725557	0.30531732	0.024013722	0.19725557

CR	0.089193825
----	-------------

Enter the names of the criteria (Step 1)

Select the Best and the Worst (Step 2)

Enter the decision-maker's preferences (Best to others: BO vector) (Step 3)

Enter the decision-maker's preferences (Others to Worst: OW vector) (Step 4)

Use the solver add-in and the weights will automatically appear in the yellow boxes (Step 5)

Fig. 9. Source file of the excel solver, modified from <https://bestworstmethod.com>.

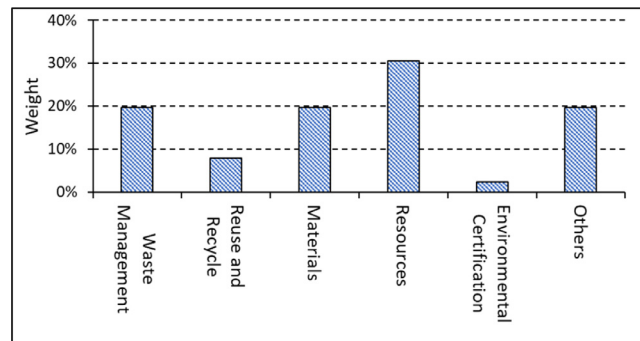


Fig. 10. The resulting BWM weights of expert #3.

In Table 14, we have shown the optimal weights of the criteria of manufacturing environmental sustainability produced by BWM. The criterion with the heaviest weight has the uppermost impact, whereas the criterion with the lowest weight has the lowest impact in evaluating manufacturing environmental sustainability. In this study, “waste management” is the criterion with the highest weight of 0.3449 which is consistent with the findings of AHP, while the “environmental certification” is the criterion with the lowest weight of 0.0312 which is not far from that found in AHP. Therefore, it is highly recommended to consider waste management as the primary concern when an organization aims to pursue manufacturing environmental sustainability. The consistency ratio obtained equals 0.0913, which is acceptable since it is close to “0” indicating a high consistency in the evaluations.

While certification falls in the least significant cluster, its importance should not be overlooked for the following reasons. First, certification is not a mandate in most countries, it is voluntarily sought only when stakeholders can see actual advantages from it. Because certification is an option rather than a requirement, the benefits will be less visible. If there are effective legislative procedures that mandates certification, the impact of certification will be noticed.

Second, while a good deal of enterprises could attain environmental certification, what matters in terms of genuine effect is business conduct and accountability, as well as regulations backed by legislation. According to literature, instances of large firms who have been certified for a long time have committed infractions of greenwashing (Alfonso et al., 2017).

Table 15

Weights of criteria in the course of sensitivity analysis.

Criteria	Original weights	Weights								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Waste management	0.3980	0.1000	0.2000	0.3000	0.4000	0.5000	0.6000	0.7000	0.8000	0.9000
Resources	0.2960	0.4425	0.3934	0.3442	0.2950	0.2458	0.1967	0.1475	0.0983	0.0492
Materials	0.1480	0.2213	0.1967	0.1721	0.1475	0.1229	0.0983	0.0738	0.0492	0.0246
Reuse and recycle	0.0800	0.1196	0.1063	0.0930	0.0797	0.0664	0.0532	0.0399	0.0266	0.0133
Environmental certification	0.0470	0.0703	0.0625	0.0547	0.0468	0.0390	0.0312	0.0234	0.0156	0.0078
Others	0.0300	0.0449	0.0399	0.0349	0.0299	0.0249	0.0199	0.0150	0.0100	0.0050
Multiplier		1.4950	1.3289	0.1628	0.9967	0.8306	0.6645	0.4983	0.3322	0.1661

Table 16

Resulting ranks of the criteria upon conducting the sensitivity analysis.

Criteria	Original rank	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Waste management	1	4	2	2	1	1	1	1	1	1
Resources	2	1	1	1	2	2	2	2	2	2
Materials	3	2	3	3	3	3	3	3	3	3
Reuse and recycle	4	3	4	4	4	4	4	4	4	4
Environmental certification	5	5	5	5	5	5	5	5	5	5
Others	6	6	6	6	6	6	6	6	6	6

Beyond the certification document, firms should have a mechanism in place to incorporate social, environmental, ethical human rights, and consumer issues into their business operations to completely satisfy their social duty.

However, with increasing national legislation such as the climate change bill (CCB) in the UK; companies may be required to implement certification like ISO14001. Lastly, while studies stated that ISO 14001 certification for the environmental management systems did improve the overall quality and environmental focus and performance, literature studies also reported the difficulty to directly link improvements to certification (Pawel et al., 2021). Clearly, if certification is obligatory, more tangible results would have been observed.

5.6. Sensitivity analysis

A sensitivity analysis, in other words “what-if analysis”, determines how elusive the output of a model while allocating different inputs. A sensitivity analysis had to be performed to ensure the robustness of the results obtained from both MCDM tools. The sensitivity analysis is conducted by changing the weight of the first-ranked criterion and prorating the weights of the other criteria (Mangla et al. 2015; Prakash and Barua, 2016).

5.6.1. Sensitivity analysis in Analytic Hierarchy Process

The sensitivity analysis for the AHP was performed by altering the average weight of “Waste Management” from 0.1 to 0.9. Using a multiplier to get the other criteria weights, a simple check can be performed on how the ranking of the criteria may change. Note that the weight of all other criteria excluding “Waste Management” in the last column in Table 12 is given by: $1 - \text{the weight of “Waste Management”} = 0.6020$. To gradually change the weight of “Waste management” while keeping the same weight proportions of the remaining criteria, the weights of all other criteria should be scaled by the multiplier $(w_j) = \frac{w_j}{1 - \sum_{i \neq j} w_i}$. For instance, when a weight of 0.1 is assigned to “Waste management”, the multiplier $m(0.1) = \frac{0.1}{0.6020} = 1.4950$, which should be used to scale the weights of all criteria excluding “waste management”.

In the next step, the factor m is then multiplied by the original weight of all other criteria to find the new weights, which are then used to rank the criteria. Table 15 is completed by applying the same steps for the weights from 0.2 to 0.9. The resulting ranks of the criteria for every new weight are shown in Table 16.

As shown in Fig. 11, except when the weight of “Waste Management” is below 0.3, its rank remains in the first place. Similarly, and even more clearly, the “others” criterion remains the last regardless the new assigned weights. This indicates the robustness of the ratings and the consistency in the evaluations. This is because the ranks do not significantly change upon making changes in the weights. Almost all the criteria remained in their positions except when the weight of “Waste Management” is low where in this case another criteria will take over the first rank.

5.6.2. Sensitivity analysis of Best Worst Method

Following a similar procedure in the sensitivity analysis of AHP, the sensitivity analysis for the BWM is also performed by altering the average weight of “Waste Management” from 0.1 to 0.9 while using the multiplier formula to find the other criteria weights. “Waste Management” weight is incrementally raised and the resulting factor is then multiplied by the original weights of all other criteria in order to keep proportionality. The findings are shown in Table 17 and the ranks in Table 18.

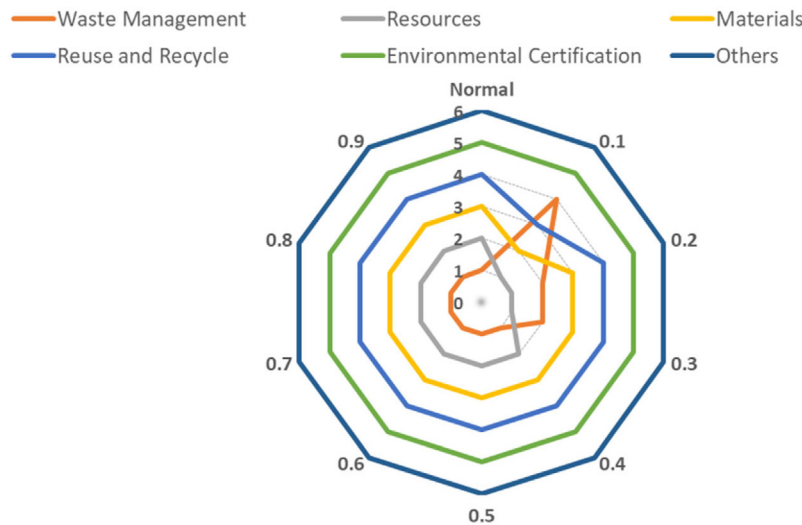


Fig. 11. A sensitive analysis by changing the weight of “Waste Management”.

Table 17

A sensitivity analysis by changing the weight of “Waste Management”.

Criteria	Original weights	New weights								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Waste management	0.3449	0.1000	0.2000	0.3000	0.4000	0.5000	0.6000	0.7000	0.8000	0.9000
Resources	0.2586	0.3553	0.3158	0.2764	0.2369	0.1974	0.1579	0.1184	0.0790	0.0395
Materials	0.1246	0.1712	0.1522	0.1332	0.1142	0.0951	0.0761	0.0571	0.0381	0.0190
Reuse and recycle	0.1102	0.1514	0.1346	0.1177	0.1009	0.0841	0.0673	0.0505	0.0336	0.0168
Others	0.1305	0.1793	0.1593	0.1394	0.1195	0.0996	0.0797	0.0598	0.0398	0.0199
Environmental certification	0.0312	0.0428	0.0381	0.0333	0.0285	0.0238	0.0190	0.0143	0.0095	0.0048
Multiplier		1.3738	1.2212	1.0685	0.9159	0.7632	0.6106	0.4579	0.3053	0.1526

Table 18

The resulting ranks of the criteria upon conducting the sensitivity analysis.

Criteria	Original rank	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Waste management	1	5	2	1	1	1	1	1	1	1
Resources	2	1	1	2	2	2	2	2	2	2
Others	3	2	3	3	3	3	3	3	3	3
Materials	4	3	4	4	4	4	4	4	4	4
Reuse and recycle	5	4	5	5	5	5	5	5	5	5
Environmental certification	6	6	6	6	6	6	6	6	6	6

Shown in Fig. 12, as the weight of “Waste Management” changes from 0.3449 to 0.1, it switches its rank from the first place to the fifth place and “Resources” jumps to the first place. However, as the weight changes to 0.2, “Waste Management” changes its rank from the fifth place to the second place while “Resources” remain in the first place. Anyway, “Waste Management” stays in the first place for the range of 0.3 till 0.9. On the other hand, “Environmental Certification” does not change its rank keeping its sixth position during the entire sensitivity analysis.

6. Results analysis, managerial and practical implication

6.1. Result analysis

After conducting the sensitivity analysis for both tools via testing different weights of the criteria, we observed insignificant changes in the ranks, where in BWM, “Waste Management” and “Resources” changed their positions only two times and then returned to their original ranks, while in AHP those criteria changed their positions three times. By averaging both results from AHP and BWM, the “Waste management” remains the most important, and “Environmental certification” becomes the least important. While Fig. 13 shows a great consistency between the results of AHP and BWM, the two approaches combined do demonstrate ranks that are more reliable. Hence, any firm that is reconsidering its

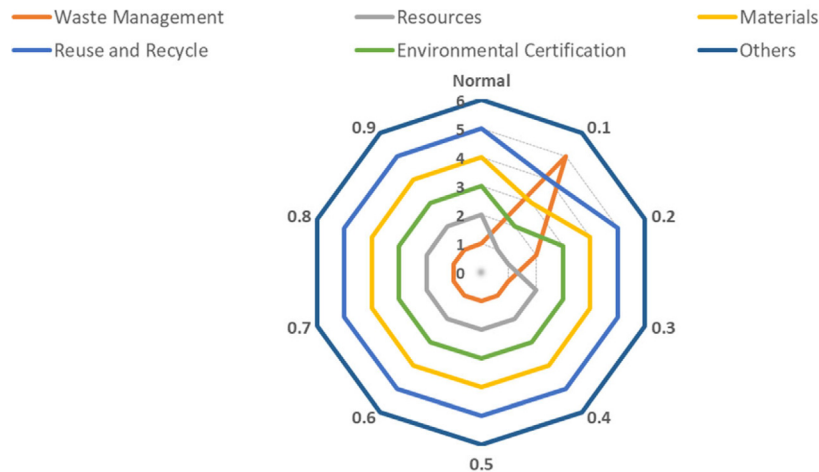


Fig. 12. The resulting ranks upon altering the weight of “Waste Management”.

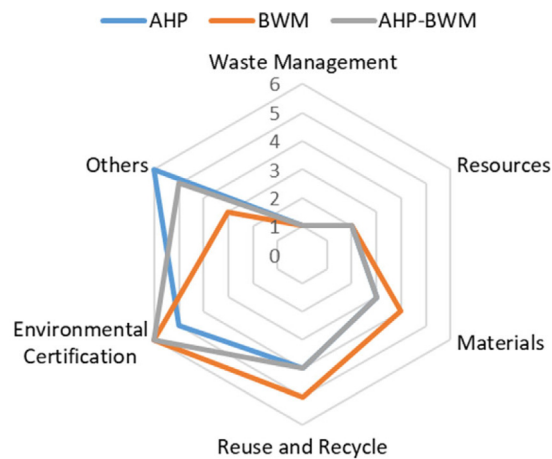


Fig. 13. AHP-BWM final ranks of each criterion (gray line).

business for the purpose of improving the sustainability standpoint should address its waste and consider the concept of design-for-recyclability.

By comparing our results with those found in some literature studies, although to the best of our knowledge, no specific study has addressed the same factors, we purposely pick (Suhi et al., 2019) who studied some comparable factors. It was found that waste management demonstrates the highest rank. This emphasizes that our results are in line with the literature with only few variations which are attributed to wood supply chains in particular as compared to other types.

6.2. Managerial and practical implications

Like any research that addresses industrial issues, it comes with its theoretical and practical implications.

Theoretical implications:

The study contributes to the theory by proposing a new model using the triplet of Hierarchical Clustering, Analytical Hierarchy Process, and Best Worst Method (HC-AHP-BWM) and argues that it will be beneficial for identifying then assessing the factors that are essential for environmental sustainability in wood supply, with UAE wood industry as a case study.

In terms of theoretical consequences, it is important to show whether our findings validate or refute a proposed hypothesis, stay in line with literature findings and can be easily comprehended by industry. Within the context of wood supply chains in UAE, the factors that are essential for environmental sustainability in wood supply chain have been identified which also match to a great deal with the literature findings. The results and sensitivity analysis demonstrated

the robustness of the model and consistency of findings. However, employing this model implies the use of optimization package and clustering software, which may not be readily available to industry.

Practical Implications:

Our study offers both practical and managerial inferences that assist organizations in policy making. First, this study helps organizations and decision makers, specifically in wooden furniture business, to focus on wood recycle and friendly waste disposal. Second, recycling excess materials such as wood chips, dust, and debris by outsourcing and contracting. Third, “design with innovation” will help increase wood utilization and reduce the waste. Lastly, the measures of environmental sustainability using the triplet of decision-making tools in this study will help organizations to achieve overall sustainability. Besides, for a successful implementation of sustainable operations, crafting product and process innovative ideas should be emphasized along within the concept of industry 4.0 as it significantly affect the development of the corporate sector. It is also stated that, rather than depending on good intentions or responding to external stimuli, organizations should deliberately recognize and use the competitive advantages they have built by being ecologically sustainable and socially conscious. Environmental considerations are crucial aspects executives are currently looking for when designing their business plans. Consumer concerns about environmental repercussions have prompted enterprises to adopt current eco-efficient manufacturing models in order to improve economic performance and profitability of process operations, as well as to develop an eco-efficient production model.

Industry 4.0 emphasizes the coherent digitization and interconnection of all productive units in an integrated environment. Therefore technological areas should encompass horizontal expansions of Industry 4.0 to include environmental matters. The core feature of the sustainable solutions, though, is the opportunity – presented by the ‘smart factory’ – of monitoring and evaluating the life cycle of every commodity in a candid and detailed way, both inside and outside the manufacturing industry. Clearly, the transformation to sustainable supply chains and processes is a gradual process which requires a certain level of reengineering. Having said that, in the process of transformation to accommodate sustainability, we underpin the following challenges:

- The absence of coherence in the policies that may not promote economic and environmental effectiveness (e.g., tariffs, subsidies, capital write-offs, and trade deals, etc.)
- Differences in cultural and linguistic approaches to collaborative processes.
- Differences in management styles and cultures.
- Poor private sector engagement in public decision formation and poor co-operation between corporate and non-governmental entities.
- Proper channels to communicate with stakeholder for tangible involvement
- Lack of initiatives that put future stakeholders through industries and diverse priorities.

7. Conclusion

Due to the absence of tangible evidences of the advances made by the Middle East region towards sustainability in manufacturing, particularly in wood industry, this study is presented to somewhat fill this gap by gathering data from stakeholders in wood industry in the region and introducing an analysis to identify and rank wood manufacturing sustainability criteria. A triplet of decision-making tools is hybridized to establish a single framework that is capable of identifying then highlighting the significance of each factor that affects sustainability in wood industries. The model begins by data reduction via hierarchical clustering, followed by an AHP analysis and lastly an optimization model of BWM. The criteria were collected from different articles and experts from the manufacturing field who are involved in environmental sustainability. The criteria were validated by multiple rounds of the selected experts. Minitab, Expert Choice and an Optimization solver were used to analyze the data. The collected factors were clustered into six main groups, namely, waste management, resources, reuse and recycle, materials, environmental certification, and others.

The results obtained from both AHP and BWM shows a remarkable consistency indicating that the most important critical criterion is “Waste Management”, and on the other hand, the least important criteria were “Others, e.g., government regulations and services, budget, etc”. and “Environmental Certification”. With this result, we conclude that not the budget nor the government regulations do slow the efforts towards sustainability in the region, in fact it is the industrial practices and waste management that need to be revisited and emphasized.

Recommendations

Our recommendation for wooden furniture industries is to inspire more sustainable operations and focus on wood recycle and friendly waste disposal. Particularly, the following may be considered as strategic actions that help wood business in the Middle East to promote manufacturing environmental sustainability, thereby, keeping the Middle East industry in the market:

- Making a competitive advantage from on-site processing of the wood waste to brand high-value goods and direct sales to the end consumer; in specific, addressing small markets.
- Take advantage of the supporting governmental regulations and good economic position in the Middle East, particularly the Gulf region.

- Recycle excess materials, wood chips, dust, and debris by outsourcing and contracting with disposal and recycle companies that take care of the residues.
- Resources came in the second place in criticality in both MCDM tools. However, a high potential exists in the Middle East to invest in solar energy, hence, more efforts should be directed to replace non-renewable resources by renewable ones to extend the life of fossil fuel and reduce emissions.
- Design for innovation will increase wood utilization and reduce the waste.
- Integrate industry 4.0 concepts with environmental policies.
- Finally, potential future work on this subject should concentrate on different geographical regions so that reliable interpretation of the inputs and outputs is feasible according to the needs and consumption habits.

The finding of this article can be used for benchmarking of the measures of environmental sustainability using the triplet of decision-making tools. Moreover, this analysis may encourage industries to improve their environmental practices and competitiveness in local and global markets. Sustainability is a home of competitive advantage in today's market. With the advent of continuing focus on environmental issues, this study comes to provide key insights for future studies in the field of environmental, social and economic sustainability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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